ParentSync

Your family's WhatsApp and email messages, automatically organized into calendar events

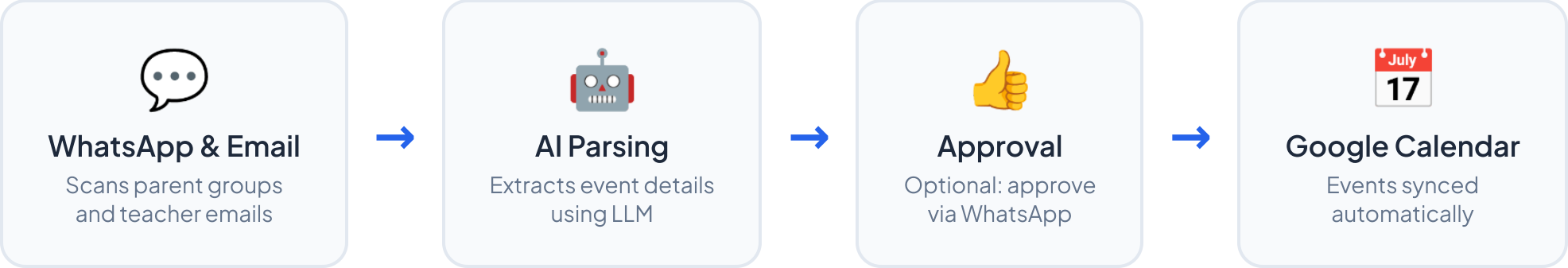
v1.0.1 | Desktop App for Linux, Windows & macOS

What is ParentSync?

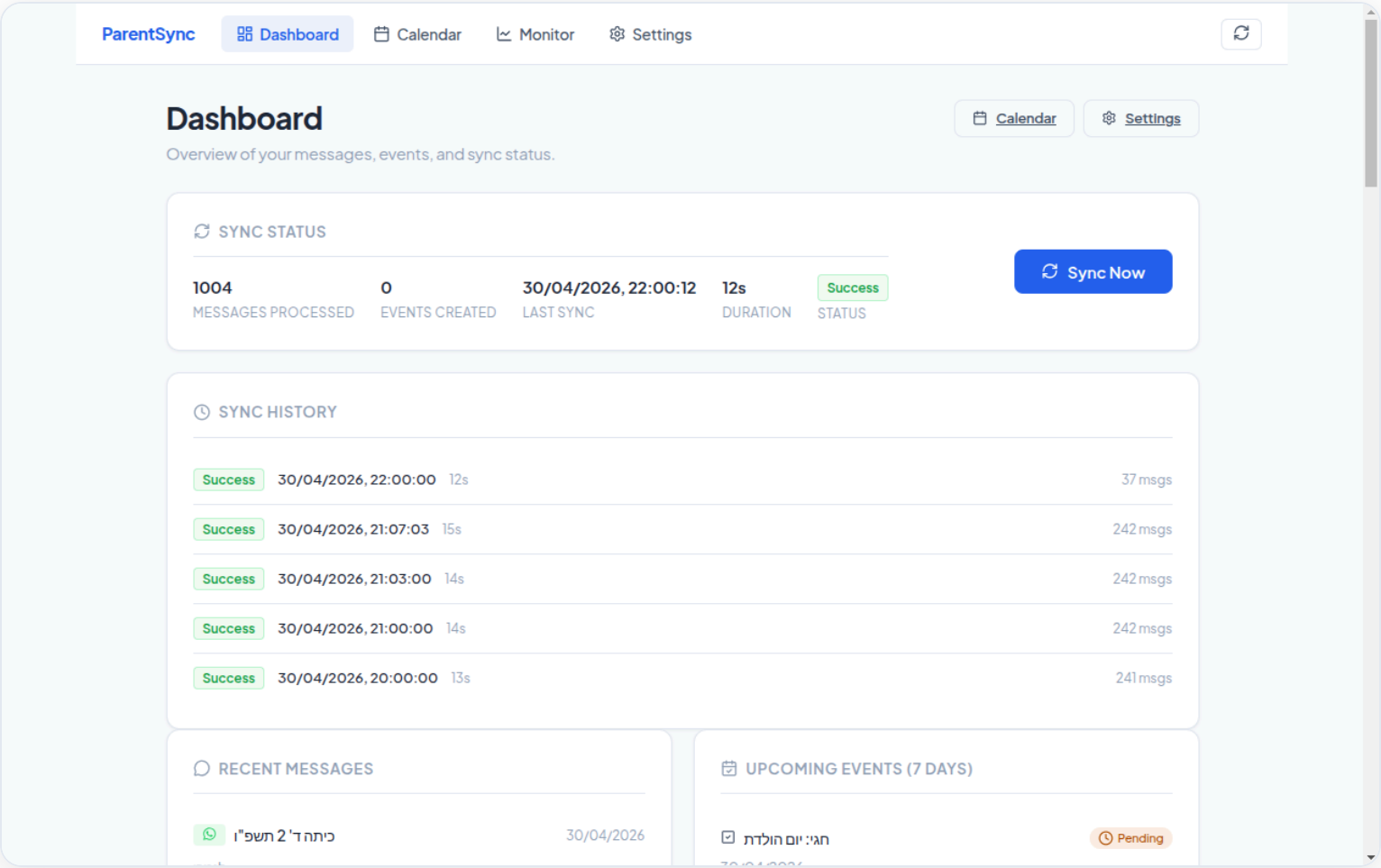
A desktop app that reads your children's school WhatsApp groups and teacher emails, uses AI to find events, and puts them on your Google Calendar.

- Monitors WhatsApp parent groups and Gmail automatically
- AI-powered parsing extracts dates, times, and event details
- Events sync directly to your family Google Calendar
- Optional approval workflow via WhatsApp reactions (👍 approve / ☹️ reject)
- All data stored locally on your machine — no cloud, no sign-up

How It Works

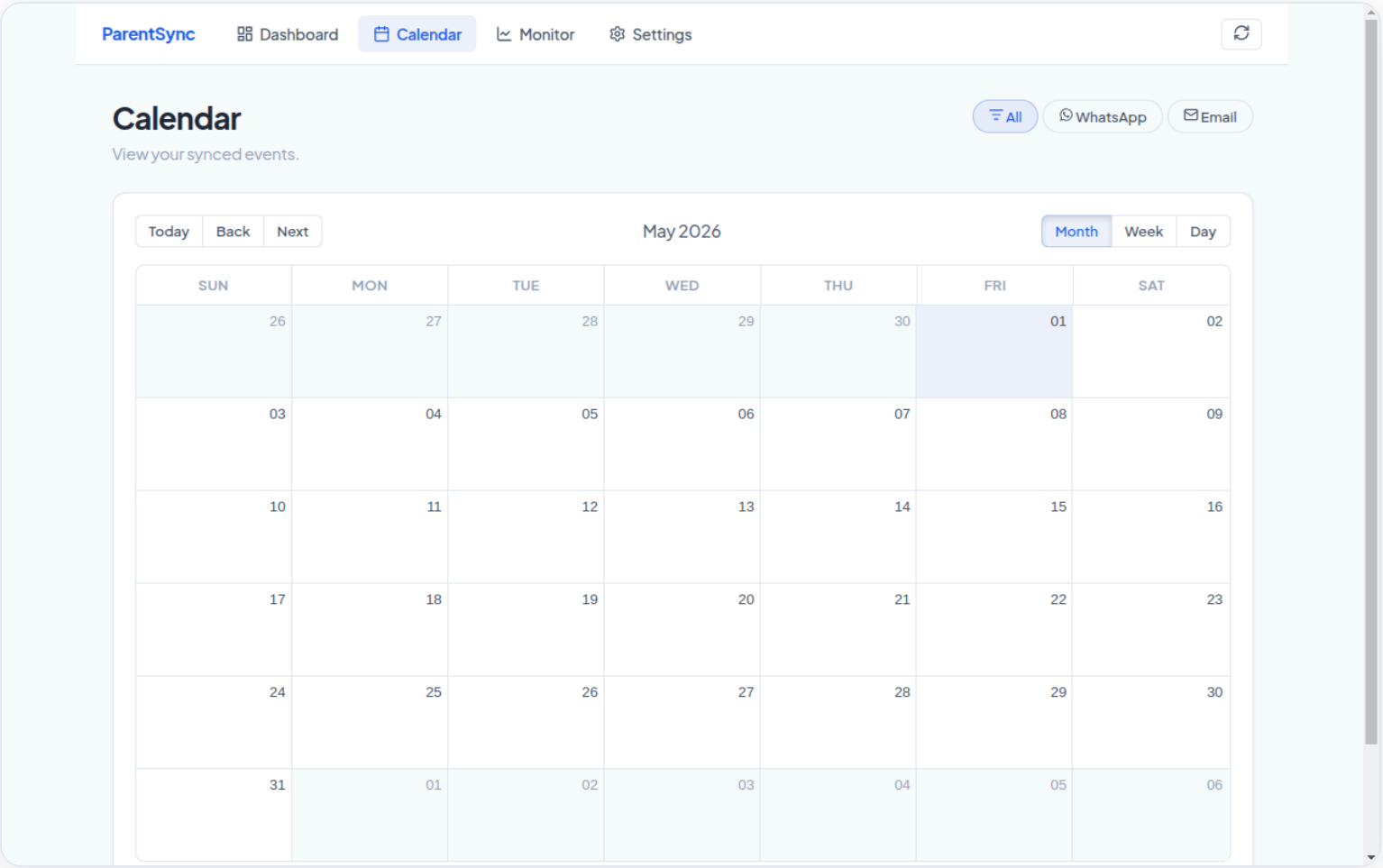

WhatsApp & Email
Scans parent groups and teacher emails
AI Parsing
Extracts event details using LLM
Approval
Optional: approve via WhatsApp
Google Calendar
Events synced automatically

Dashboard



Sync status, recent messages, upcoming events, and sync history at a glance

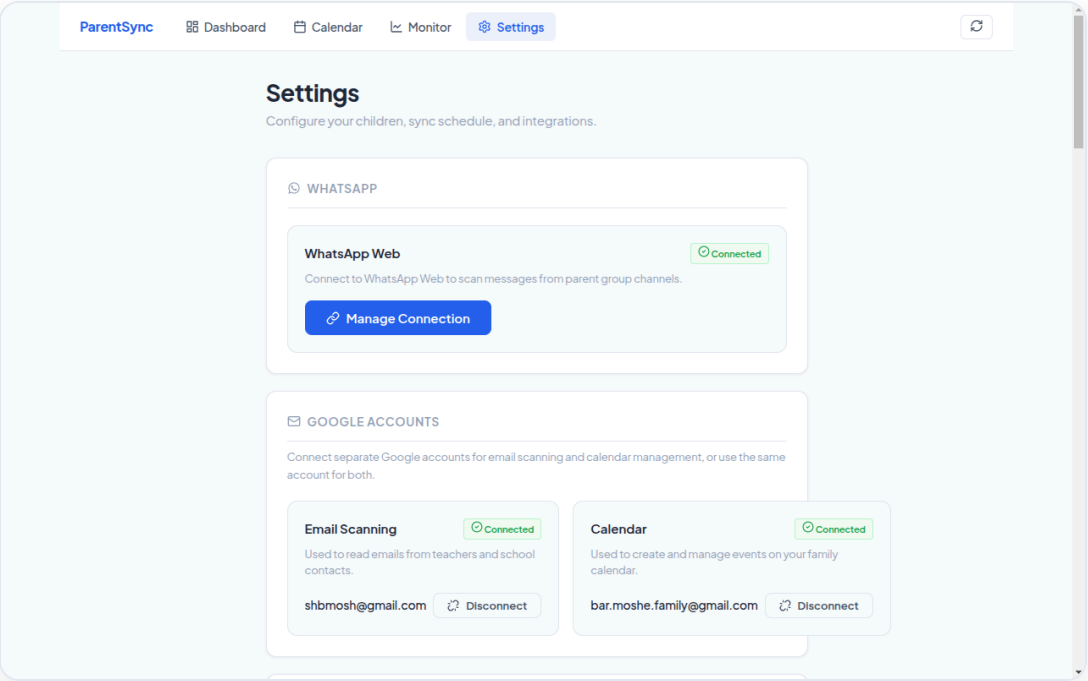
Calendar View



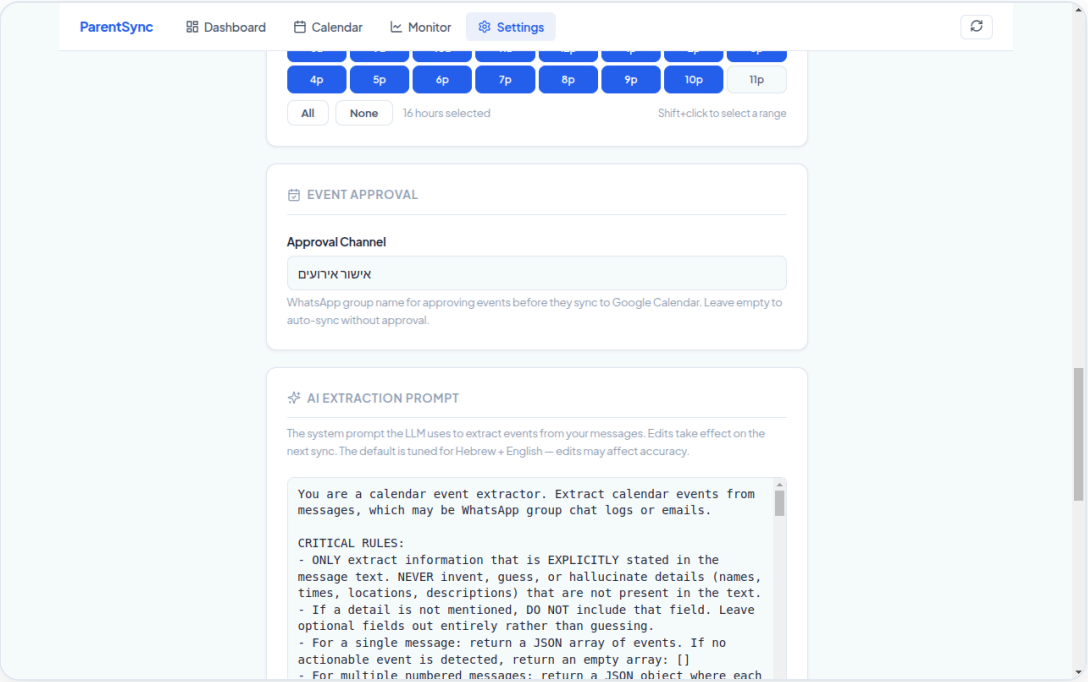
Full month/week/day calendar with events color-coded by source (WhatsApp or Email)

Settings & Setup

Connections



Children & Schedule



Dashboard — In Detail

Sync Status (top card)

- **Messages Processed** — total messages collected across all recent syncs
- **Events Created** — how many calendar events the AI extracted
- **Last Sync** — when the last sync happened
- **Duration** — how long it took (e.g., 18s)
- **Status badge** — green "Success", orange "Partial", or red "Failed"

💡 Click **"Sync Now"** to trigger an immediate scan of all your configured WhatsApp groups and email addresses.

Sync History (expandable list)

- Shows the **last 5 syncs** with status, time, duration, message count
- **Click any entry** to expand — see per-channel breakdown (which child, which group, how many messages)
- Skipped channels show the **reason** (e.g., "no new messages")
- Click **"View synced messages"** to see exact messages that were collected

Dashboard — Messages & Events

Recent Messages (bottom left)

- All messages collected from WhatsApp and Gmail
- Source icon: **WhatsApp icon** = WhatsApp group, **Mail icon** = email
- **Click a message** to expand and read full content
- Click **"Show more"** to load additional messages (20 at a time)

Upcoming Events (bottom right)

- Events in the **next 7 days** parsed from messages
- Coloured pill: 🟠 **Pending**, 🟢 **Approved**, 🔴 **Rejected**
- Pending events have inline **Approve / Reject** buttons — same effect as 👍 / 😞 in WhatsApp
- **Checkmark icon** = synced to Google Calendar

How to get data flowing

1. Make sure **WhatsApp is connected** (Settings → WhatsApp → "Connected")
2. Add at least one **child with WhatsApp channels** (Settings → Children)
3. Set your **Gemini API key** (Settings → Gemini AI), or use OpenRouter as alternative
4. Click **"Sync Now"** on the Dashboard
5. Wait 15–30 seconds — messages and events appear

💡 **Refresh button** (top-right ↻ icon) reloads all data on the current page without triggering a new sync.

Getting & Refreshing Data

Automatic Sync

ParentSync automatically scans at the hours you configured in Settings → Sync Schedule. For example, if you selected 8am–10pm, it scans once per hour during those times.

Manual Sync

Click **"Sync Now"** on the Dashboard to trigger an immediate scan. Useful after adding a new child or channel.

Refresh Display

The **↻ refresh button** (top-right corner of the nav bar) reloads all data on the current page from the database. It does NOT trigger a new sync — it just refreshes the view.

Data Freshness Tips

1. After connecting WhatsApp for the first time, click **"Sync Now"** to populate initial data
2. Add all your children's group names **before** the first sync so all channels are scanned
3. The Monitor page filters (7d/30d/90d) affect all charts — use **30 days** for a good overview
4. If Dashboard shows old data, click **↻ refresh** — if it still doesn't update, click **"Sync Now"**

💡 The app keeps running and syncing even when minimized to the system tray.

How-To Guide

Step-by-step setup for each integration

Setup: Gemini (default AI)

The AI engine that reads messages and extracts calendar events

What is Gemini?

Google's Gemini family of LLMs — fast, cheap, and very good at multilingual extraction. ParentSync uses it to read WhatsApp and email messages and identify dates, events, and details.

💡 The default model `gemini-2.5-flash-lite` is fast and cheap — typical extraction costs are negligible. A free tier is available too.

🔁 Don't want Gemini? **OpenRouter** is supported as an alternative provider — see next slide.

How to get your API key

1. Go to **aistudio.google.com/apikey**
2. Sign in with your Google account
3. Click **"Create API key"**
4. Copy the key
5. Paste in ParentSync → Settings → Gemini AI → API Key
6. Optionally pick a different model in the same section

📺 **Video walkthrough** — Gemini API Key in Google AI Studio (YouTube)

Alternative: OpenRouter

One-stop shop for many LLM providers behind a single API key

What is OpenRouter?

A gateway to many AI models (GPT, Claude, Llama, Qwen, etc.) through a single API key. Useful if you already have credits there, or if you want to A/B test models without changing keys.

💡 Free models are available — typically labelled `:free`. They have rate limits but cost nothing.

How to get your API key

1. Go to **openrouter.ai/keys**
2. Sign up or log in (Google/GitHub)
3. Click **"Create Key"**
4. Copy the key (starts with `sk-or-`)
5. Paste in ParentSync → Settings → OpenRouter → API Key

📺 **Video walkthrough** — Get your OpenRouter API Key (YouTube)

You can fill both keys; the active provider is whichever the build is wired to (Gemini by default).

Setup: Google Account

Connect Gmail and Google Calendar to read emails and sync events

What is Google OAuth?

OAuth lets ParentSync access your Gmail and Calendar **without storing your Google password**. You create a "Client ID" in Google Cloud Console — think of it as a permission slip that tells Google "this app is allowed to access my data."

💡 You can use **separate Google accounts** for email scanning and calendar — or the same account for both.

How to set it up

1. Go to **console.cloud.google.com**
   Create a project if you don't have one
2. Enable **Gmail API** and **Calendar API**
   APIs & Services → Enable APIs
3. Create **OAuth consent screen**
   Add your email as a Test user
4. Create **OAuth Client ID** (Web app)
   Redirect URI:
   `http://localhost:41932/api/auth/google/callback`
5. Copy **Client ID** and **Client Secret** into ParentSync Settings

Connecting Google Accounts

After entering Client ID & Secret

1. Click **"Save Settings"**
2. Click **"Sign in with Google"** under Email Scanning
3. Google's consent screen opens in your browser
4. Select your account and click **"Allow"**
5. You're redirected back — status shows **Connected**
6. Repeat for **Calendar** (same or different account)

Troubleshooting

⚠️ **"Error 403: access_denied"**

Your Google app is in "Testing" mode. Fix: Go to OAuth consent screen → either **add your email as a Test user** or click **"Publish App"**.

⚠️ **"Failed to exchange authorization code"**

The Client Secret may be wrong, or the redirect URI doesn't match. Double-check both in Google Cloud Console and in ParentSync Settings.

Setup: WhatsApp

Connect your WhatsApp to scan parent group messages

How it works

ParentSync connects to WhatsApp Web — the same way you'd use WhatsApp on your computer. It runs its own browser in the background to read messages from the groups you specify.

💡 Your WhatsApp session **persists across app restarts**. You only need to scan the QR code once. If the session expires, you'll be prompted to re-scan.

📱 Your phone must stay connected to the internet for WhatsApp Web to work (WhatsApp requirement).

How to connect

1. Open ParentSync → **Settings**
2. Click **"Connect WhatsApp"**
3. A **QR code** appears in the app
4. On your phone:
   WhatsApp → Settings → Linked Devices → Link a Device
5. Scan the QR code with your phone camera
6. Status changes to **"Connected"**

Then go to **Children** in Settings, add a child, and type the exact WhatsApp group name.

Under the Hood

DESKTOP SHELL
Electron (single executable)

BACKEND
NestJS (REST API, embedded)

AI
Gemini (default) · OpenRouter swap-in

FRONTEND
React + TypeScript

DATABASE
SQLite (local, no server)

WHATSAPP
whatsapp-web.js (Chromium)

All data stored locally on your machine. No cloud servers, no accounts to create. Just download, configure, and run.

Quick Start — Get the App

Click your platform — direct download from GitHub Releases v1.0.2. Per-platform run instructions on the next slides.

🍎

macOS

- ⬇️ Apple Silicon arm64.dmg
- ⬇️ Intel .dmg

🪟

Windows 10/11

- ⬇️ Installer Setup-1.0.2.exe

🐧

Linux

- ⬇️ Portable .AppImage
- ⬇️ Debian/Ubuntu .deb

No account needed. All assets: github.com/ShaharBarMoshe/ParentSync/releases/tag/v1.0.2

Run on macOS 🍎

⬇️ Download — Apple Silicon ParentSync-1.0.2-arm64.dmg

⬇️ Download — Intel ParentSync-1.0.2.dmg

Step by step

1. Click the right download above (M-series Mac → Apple Silicon; older Intel Mac → Intel)
2. **Double-click** the `.dmg` in your Downloads folder
3. **Drag** the ParentSync icon into the **Applications** folder
4. Open **Applications** and double-click ParentSync
5. First time only: macOS says "Apple cannot check it for malicious software."
   Right-click the icon → **Open** → **Open** in the dialog

What you'll see

- The app window opens — click **Settings** (top bar) to start configuring
- An icon appears in your menu bar (top right) — right-click for Sync Now / Quit
- Closing the window keeps it running; quit fully from the menu bar

📦 The build is unsigned (no Apple Developer certificate). The right-click → Open dance is only needed the first time.

Run on Windows 🪟

⬇️ Download Installer ParentSync-Setup-1.0.2.exe

Step by step

1. Click the download button above
2. **Double-click** the installer in your Downloads folder
3. Windows SmartScreen says "Windows protected your PC."
   Click **"More info"** → **"Run anyway"**
4. Click through the installer (Next → Install → Finish)
5. The app launches automatically and adds itself to the Start menu

What you'll see

- A desktop shortcut and a Start-menu entry are created
- An icon appears in the system tray (bottom right) — right-click for Sync Now / Quit
- Closing the window minimizes to the tray; right-click → Quit to exit fully
- Auto-starts on login by default

📦 The installer is unsigned (no Authenticode certificate). The SmartScreen "Run anyway" is only needed the first time.

Run on Linux 🐧

⬇️ AppImage (any distro) ParentSync-1.0.2.AppImage

⬇️ .deb (Debian/Ubuntu) parentsync_1.0.2_amd64.deb

AppImage (any distro)

1. Click the AppImage download above
2. Files need to be marked runnable first. Two ways:
   Mouse: Right-click → **Properties** → **Permissions** → check **"Allow executing as a program"**.
   Terminal: `chmod +x ParentSync-1.0.2.AppImage`
3. **Double-click** it (or `./ParentSync-1.0.2.AppImage`)

Debian / Ubuntu: .deb

1. Click the .deb download above
2. Double-click → opens in Software Center → **Install**
3. Or terminal: `sudo dpkg -i parentsync_1.0.2_amd64.deb`
4. Launch from your application menu

Auto-start on login (optional)

Clone the repo and run `npm run install:local` — registers the AppImage as a systemd user service.

Platform Support & Building From Source

Platform	Format	How to Run
Linux	.AppImage (portable)	chmod +x, double-click or run from terminal
Linux	.deb (Debian/Ubuntu)	dpkg -i, run from app menu
Windows	.exe (NSIS installer)	Double-click installer
macOS	.dmg	Drag to Applications

```
# Build it yourself (any platform)
git clone https://github.com/ShaharBarMoshe/ParentSync.git
cd ParentSync && ./setup.sh
npm run package:linux # or :mac / :win
```

Built with Electron | All data stored locally | Private use

Uninstalling 🗑️

An in-app **Settings → Danger Zone → Uninstall ParentSync** button is planned (Phase 19) — until then, here's the manual cleanup per platform. Add `--purge` / the data-removal step to also wipe your database, OAuth tokens, WhatsApp session, and logs.

🐧 Linux

```
# 1. systemd autostart
systemctl --user stop parentsync.service
systemctl --user disable parentsync.service
rm -f
~/.config/systemd/user/parentsync.service

# 2. binary + versions
rm -f ~/.local/bin/ParentSync.AppImage
rm -rf ~/.local/share/parentsync
rm -f
~/.local/share/applications/parentsync.desktop

# 3. (.deb only)
sudo apt remove parentsync

# 4. data — IRREVERSIBLE
rm -rf ~/.config/parentsync
```

🍎 macOS

```
# 1. Quit ParentSync (menu bar)
# 2. Drag /Applications/ParentSync.app to Trash

# 3. data — IRREVERSIBLE
rm -rf "$HOME/Library/Application Support/ParentSync"
rm -rf "$HOME/Library/Logs/ParentSync"
rm -rf "$HOME/Library/Caches/com.parentsync.app"
rm -f
"$HOME/Library/Preferences/com.parentsync.app.plist"
rm -f
"$HOME/Library/LaunchAgents/com.parentsync.app.plist"
```

🪟 Windows

```
# 1. Settings → Apps → ParentSync → Uninstall
# (Handles binary, Start menu, registry)

# 2. data — IRREVERSIBLE (PowerShell)
Remove-Item -Recurse -Force "$env:APPDATA\ParentSync
Remove-Item -Recurse -Force
"$env:LOCALAPPDATA\ParentSync"
Remove-ItemProperty
"HKCU:\Software\Microsoft\Windows\CurrentVersion\Run
"ParentSync"
```

📱 After removing user data, open WhatsApp on your phone → **Settings → Linked Devices** and remove the ParentSync entry to fully unlink.

Event Reminders

ParentSync sends a WhatsApp reminder ~24 hours before an event begins, but only for events that have settled in the system long enough to be trusted.

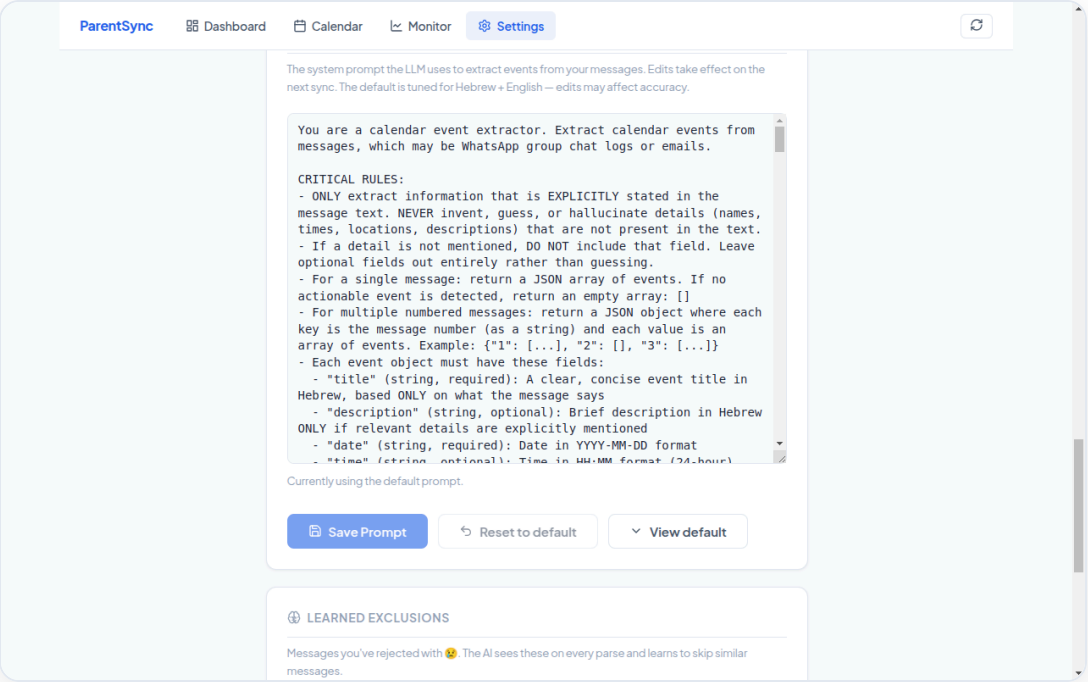
- Hourly cron scans the calendar for upcoming events
- Only events added more than **24 hours ago** are eligible — protects against last-minute parsing mistakes
- Only events already **synced to Google Calendar** qualify
- Event existence is **verified against Google Calendar** before sending — deleted events are silently skipped
- Reminder is delivered to the configured approval/notification WhatsApp chat with full details: title, date, time, location, description, and source channel
- Each event is reminded at most once (`reminderSent` flag)

See `docs/EVENT-REMINDERS.md` for the full specification.

You're in control of the AI 🎛️

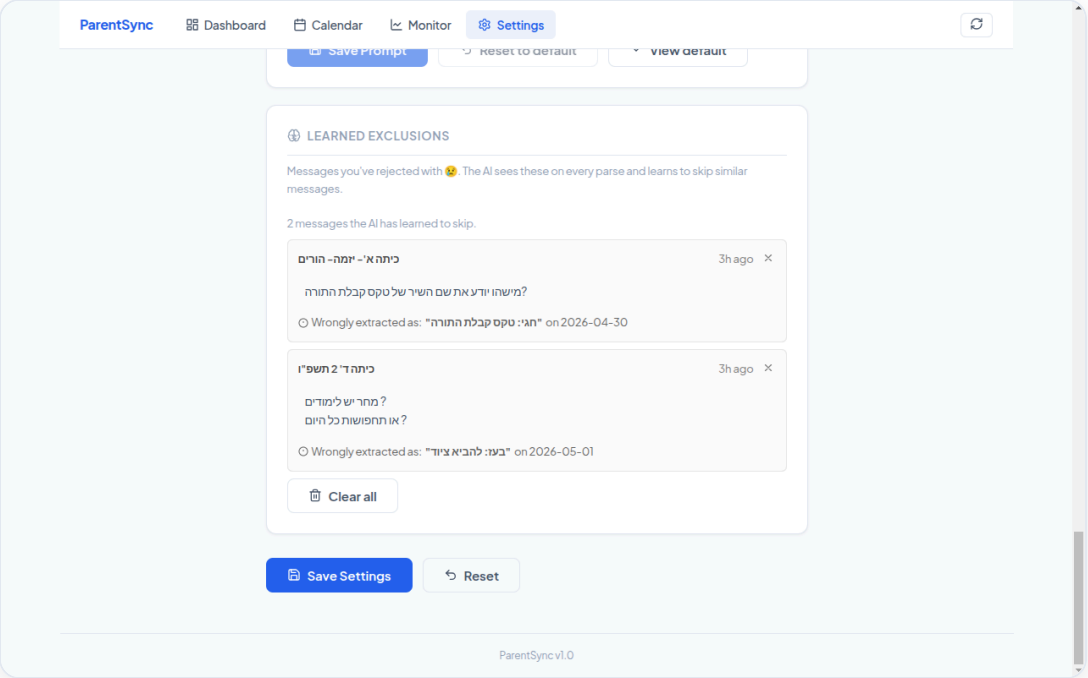
The AI never gets the last word — **you do**, in two complementary ways. The prompt sets the rules; your reactions tune the edge cases. Both feed into the same LLM call, and changes take effect on the next sync.

1. Edit the rules



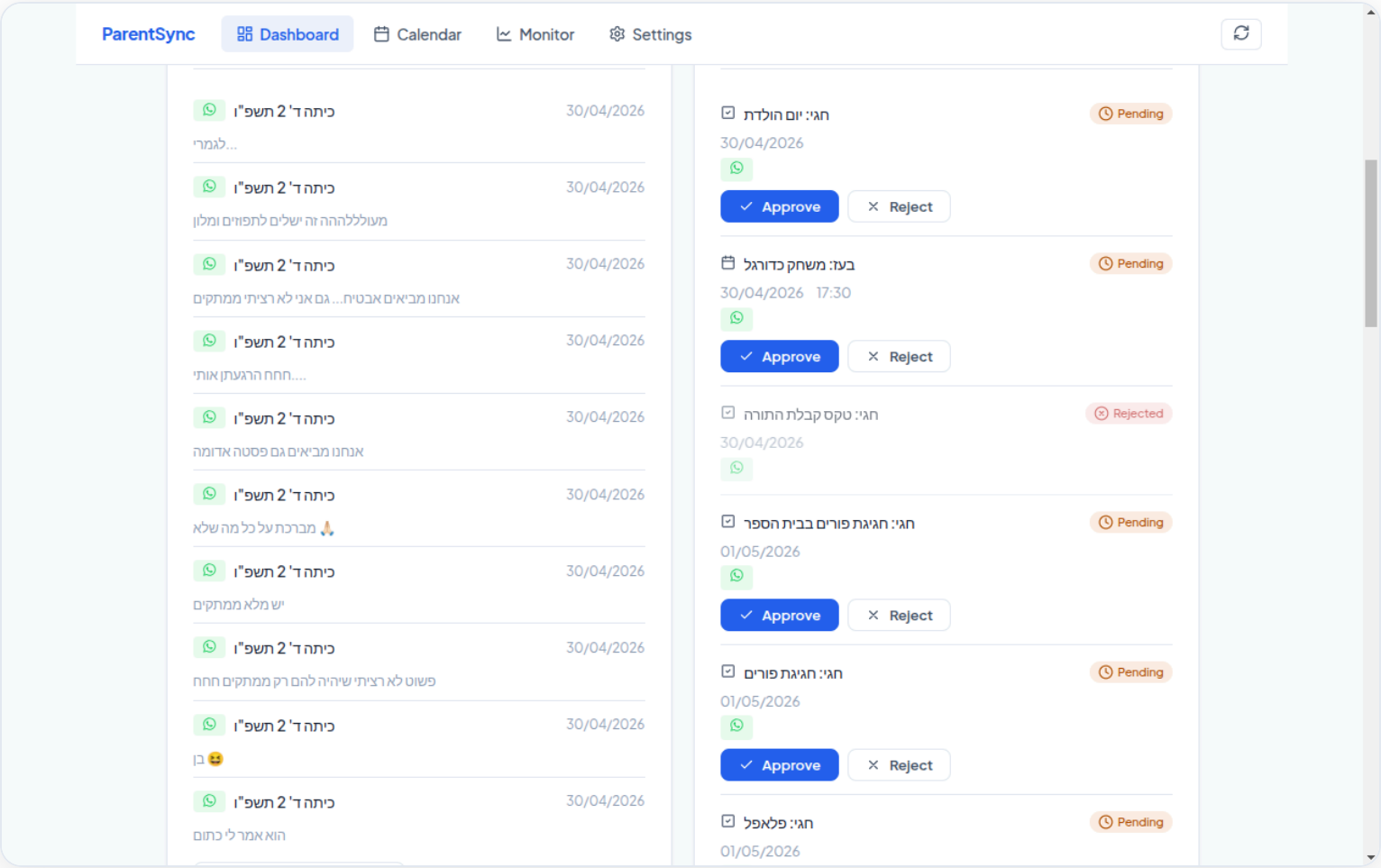
Settings → AI Extraction Prompt. The whole system prompt is a textarea. Add an example, change wording, or hit *Reset to default*. Effective on the next sync.

2. React to the results



👍 / ☹ **in WhatsApp** — or the inline buttons on the Dashboard. Every ☹ captures the source as a *Learned Exclusion* (most recent 50 appended to the prompt). Take back a reaction to undo.

In-App Approval



Each upcoming event shows its status as a pill — 🟠 Pending, 🟢 Approved, 🔴 Rejected. Pending events have inline Approve / Reject buttons. Same effect as 👍 / ☹️ in WhatsApp; reactions are reversible too — take back a 👍 to unsync from Google, take back a ☹️ to clear the learned exclusion.

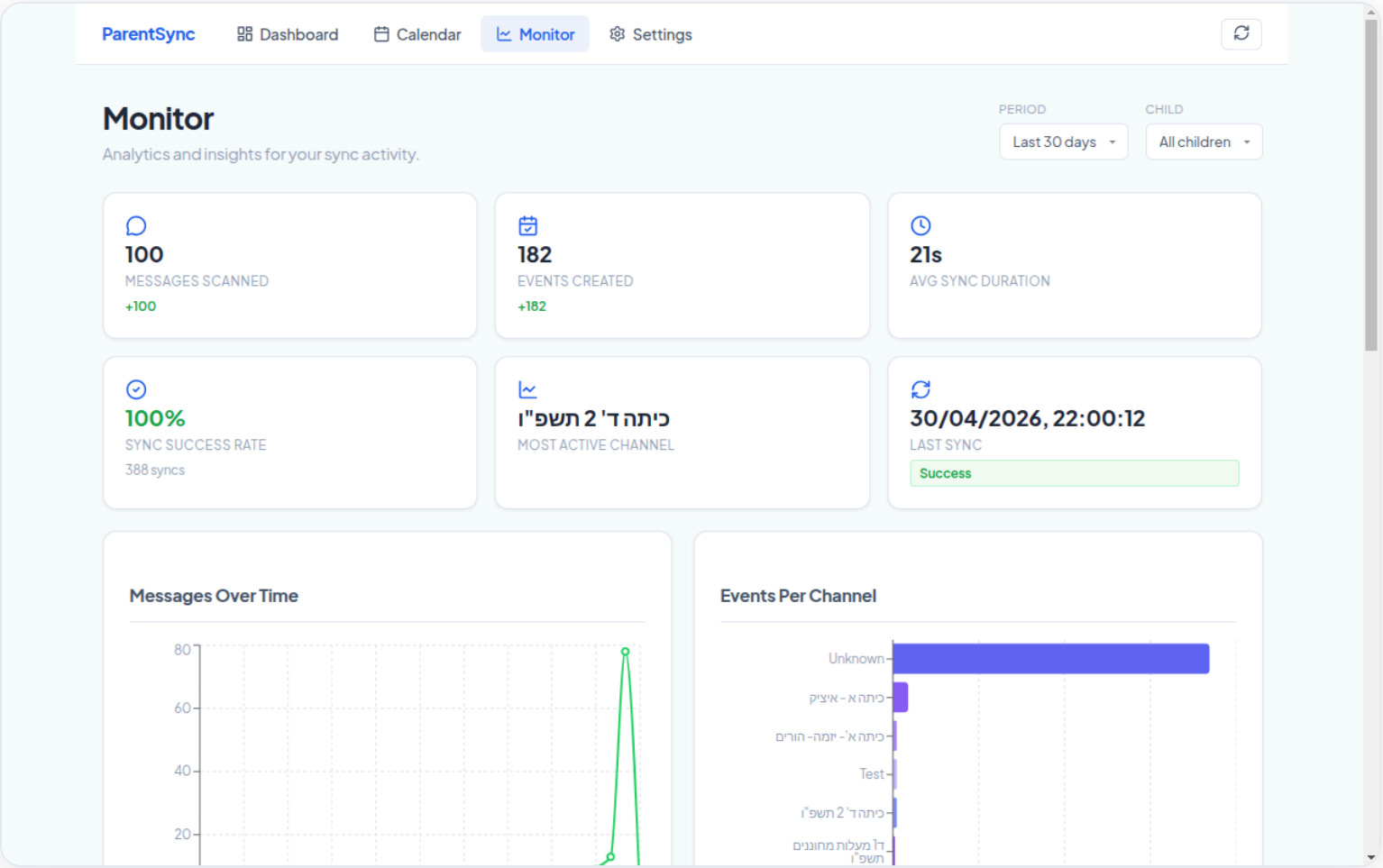
Duplicate Suppression

One real-world WhatsApp message can describe the same gathering from multiple angles — a birthday is also a "meetup," a class trip is also a "ceremony." The LLM used to extract them as *separate* events at the same date+time. Now:

- Before sending an event for approval, the backend looks up other non-rejected events for the same child at the same date+time slot.
- For each match, a focused LLM call asks **"are these two events the same gathering?"** The prompt is tiny and cheap.
- If the answer is yes, the new event is silently rejected and never reaches your approval channel — the existing event remains the canonical record.
- Date-only tasks (no time) skip the check entirely — the false-positive risk is too high without a time slot.
- LLM hiccups default to "different" — a transient error never accidentally swallows a real event.

💡 The check pairs with the negative-feedback loop: even if a duplicate slips through, one ☹ reaction puts it in the learned-exclusions pool so it's gone for good.

Monitor & Analytics



Messages scanned, events created, sync success rate, and channel activity charts

Monitor — Summary Cards

Six cards at the top give you a quick health check. Use the **Period** dropdown (7/30/90 days) and **Child** filter to narrow the view.

💬 Messages Scanned

Total WhatsApp + email messages collected. Green trend arrow shows change vs. previous period.

⏱️ Avg Sync Duration

Average time per sync cycle (scanning + parsing). Under 30s is normal.

📊 Most Active Channel

The WhatsApp group or email source that generated the most messages in this period.

📅 Events Created

How many calendar events the AI extracted from messages. Trend arrow compares to previous period.

✅ Sync Success Rate

Percentage of syncs that completed without errors. Green ≥80%, orange ≥50%, red below 50%.

🔄 Last Sync

Timestamp and status badge of the most recent sync. Tells you when data was last refreshed.

Monitor — Charts & Graphs

📈 Messages Over Time

Line chart with two lines:

🟢 **Green** = WhatsApp messages

🔴 **Red** = Email messages

X-axis: dates, Y-axis: message count. Shows daily volume trends — helps spot when groups are most active.

📊 Events Per Channel

Horizontal bar chart showing event count by WhatsApp group name. Each channel gets a different purple shade. Hover to see percentage of total. Identifies which groups generate the most calendar events.

📉 Sync History

Combo chart — bars + line overlay.

Bars: message count per sync (green = success, orange = partial, red = failed)

Line: sync duration in seconds (right axis)

Spot slow syncs or recurring failures at a glance.

🗓️ Channel Activity Heatmap

Grid with channels as rows, dates as columns. Cell color intensity = message count. Darker purple = more messages. Hover any cell for exact count. Reveals which days and channels are busiest.

Common Errors & Solutions

⚠️ "Sync failed"

WhatsApp may be disconnected, or the LLM API key is missing/expired. Check Settings → WhatsApp status and Gemini / OpenRouter API key.

⚠️ "0 Events Created" after sync

The AI didn't find any events in the messages. This is normal if messages are just chat. Only messages with dates/times get parsed into events.

⚠️ "Google OAuth not configured"

You haven't entered the Google Client ID and Secret yet. Go to Settings → Google OAuth and follow the setup steps (slide 17).

⚠️ "WhatsApp is not connected"

Session expired. Go to Settings → click "Connect WhatsApp" → scan QR code again with your phone.

⚠️ "Failed to load dashboard data"

Backend may still be starting. Wait a few seconds and click the **refresh button** (↻ top-right). If it persists, restart the app.

⚠️ Events not appearing on Google Calendar

Check: 1) Google Calendar account is connected (Settings), 2) If using approval channel, react 👍 to approve the event in WhatsApp.